

Title

FAERS Pharmacovigilance Signal Detection

Vedolizumab Adverse Event Signal Analysis

FDA Adverse Event Reporting System (FAERS) — 2025 Q1

Prepared by:

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Project Type:

Healthcare Data Analytics Project

Tools Used:

SQL | SQLite | Tableau Public | Excel | GitHub

Project Overview

Pharmacovigilance plays an important role in detecting potential safety concerns associated with medications after market approval. This project explored adverse event reporting patterns associated with **Vedolizumab** using data obtained from the **FDA Adverse Event Reporting System (FAERS) 2025 Q1 subset**.

The analysis focused on identifying disproportionality signals through **Reporting Odds Ratio (ROR)** and **Proportional Reporting Ratio (PRR)** calculations to determine whether certain adverse reactions were reported disproportionately among Vedolizumab-associated cases.

Objective

The objective of this project was to develop a reproducible healthcare analytics workflow for pharmacovigilance signal detection by:

- Cleaning and preprocessing FAERS data
- Removing duplicate adverse event reports
- Building an integrated master dataset using SQL
- Identifying drug–reaction associations
- Calculating disproportionality metrics (ROR and PRR)

- Visualizing safety signals using Tableau dashboards
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Dataset Description

Data Source:
FDA Adverse Event Reporting System (FAERS)

Quarter Analyzed:
2025 Q1

- Datasets Used:**
- DEMO (patient demographics)
 - DRUG (drug information)
 - REAC (adverse reactions)

Dataset Summary

Metric	Value
Total Reports	5,924
Total Drugs	146
Total Reactions	1,438
Highest ROR Observed	4,995

METHODOLOGY

Data Preparation Workflow

The analysis followed a structured pharmacovigilance workflow:

- FAERS Data Import**
- ↓
- Data Cleaning**
- ↓
- Deduplication of Drug and Reaction Records**

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SQL-Based Dataset Integration
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Signal Detection (Drug–Reaction Counts)
↓
ROR and PRR Calculation
↓
Dashboard Visualization in Tableau

Data Cleaning and Deduplication

To improve analytical quality:

- Only **Primary Suspect (PS)** drugs were retained to focus on drugs most likely associated with reported events.
- Duplicate drug and reaction records sharing the same **primary ID** were removed.
- Reports containing:
 - missing age,
 - age equal to zero,
 - missing sex information

were excluded from the final analytical dataset.

Additionally, indication-related and administrative reporting terms such as:

- *Crohn's disease*
- *Off label use*
- *Product dose omission issue*
- *Product use issue*

were excluded to improve interpretability of clinically relevant safety signals.

Signal Detection Methodology

Signal detection was performed using a **2×2 contingency table framework**.

Definitions

a = Reports containing the target drug and target reaction

b = Reports containing the target drug and all other reactions

c = Reports containing other drugs with the target reaction
d = Reports containing all other drugs and all other reactions

Reporting Odds Ratio (ROR)

ROR was calculated to determine whether a reaction was reported disproportionately more frequently with Vedolizumab compared to other drugs.

Formula:

$$\text{ROR} = (a \times d) / (b \times c)$$

Proportional Reporting Ratio (PRR)

PRR was calculated to compare reaction reporting proportions between Vedolizumab and comparator drugs.

Formula:

$$\text{PRR} = [a / (a+b)] \div [c / (c+d)]$$

RESULTS & KEY FINDINGS

Dashboard Analysis

A Tableau dashboard was developed to visualize:

- Key pharmacovigilance metrics
- Strongest Vedolizumab-associated signals
- ROR–PRR signal distribution
- Patient demographic trends

The dashboard included:

- KPI cards
- Top safety signal bar charts
- Signal landscape scatter plots
- Age distribution analysis

- Sex distribution analysis
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Key Safety Signals Identified

Following SQL cleaning and signal filtering, elevated disproportionality signals were observed for:

- **Diarrhoea**
- **Abdominal Pain**
- **Blood Pressure Increased**
- **Influenza**
- **Malaise**
- **Arthralgia**
- **Fatigue**

Among these, **diarrhoea** and **abdominal pain** demonstrated comparatively stronger disproportionality metrics and may warrant further pharmacovigilance attention.

The scatter plot analysis further demonstrated that reactions with higher ROR and PRR values clustered toward the upper-right quadrant, suggesting stronger disproportionality compared with background reporting patterns.

Interpretation

The findings suggest that several adverse reactions appeared disproportionately represented among Vedolizumab-associated FAERS reports within the analyzed subset. However, these signals should be interpreted as **hypothesis-generating observations rather than evidence of causality**.

FAERS-based signal detection identifies reporting patterns and potential safety concerns but does not establish direct drug causation.

LIMITATIONS & CONCLUSION

Limitations

This analysis has several limitations:

- FAERS is a **spontaneous reporting system**, making it susceptible to underreporting.
 - Reporting bias may influence signal strength.
 - Duplicate reporting and incomplete patient records may affect interpretation.
 - ROR and PRR indicate disproportionality, **not causality**.
 - The analysis was restricted to a **single FAERS quarter (2025 Q1)**.
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Conclusion

This project successfully developed a reproducible pharmacovigilance analytics workflow using **SQL, Tableau, and FAERS data** to identify potential Vedolizumab-associated adverse event signals.